AYISHATH RUSAINA

M.Tech., Water Resources Engineering, 2027

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EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	M.Tech	National Institute of Technology Calicut, Kerala	8.1 / 10
2021	B.Tech	LBS College of Engineering Kasaragod, Kerala	7.92 / 10
2016	Class XII CBSE	Crescent English School, Kasaragod, Kerala	74.4%
2014	Class X CBSE	Crescent English School, Kasaragod, Kerala	9.6 / 10

EXPERIENCE

Graduate Engineering Apprentice | **Kerala State Pollution Control Board** | **Kasaragod, Kerala** **Feb 2023 – Jan 2024**

- Assisted in water quality sampling, laboratory analysis, and gain exposure to parameters relevant to river and groundwater assessment.
- Reviewed effluent treatment units during field inspections, identifying compliance gaps in discharge standards and pollution-control measures.
- Assisted in hazardous waste, wastewater, and air-quality audits, gaining hands-on exposure to environmental monitoring standards and regulatory frameworks.
- Compiled technical notes and inspection reports, improving the clarity and accuracy of the documentation used for compliance decisions.

Engineer (via Pabsta Engineers) | **Indian Oil Corporation Ltd.** | **Kozhikode, Kerala** **Oct 2024 – Apr 2025**

- Managed engineering workflows using SAP-MM, including purchase orders, service entry sheets, and contractor invoice processing.
- Coordinated with vendors and site teams for the execution of modernization works, enhancing understanding of the infrastructure project cycles.
- Performed quantity checks, escalation calculations, and compliance verifications for civil and mechanical works.
- Supported site visits and reviewed work-permit requirements as part of project execution, improving coordination and documentation accuracy.

PROJECTS

Water Quality Analysis & WQI Evaluation for Mogral River | **B.Tech Project** | **LBSCE Kasaragod, Kerala** **Sep 2020 – Mar 2021**

- Assessed the pollution status of the Mogral River by analyzing physicochemical water quality parameters across multiple sampling locations.
- Computed the Water Quality Index (WQI) using standard methods to quantify spatial variability in river health and identify critical polluted stretches.
- Performed laboratory testing, data pre-processing, and parameter interpretation to identify contamination trends.
- Generated insights contributing to the river-monitoring effort under the National Green Tribunal (NGT) directive.

Coir-Reinforced Concrete (CRC) Experimental Study | **B.Tech Design Project** | **LBSCE Kasaragod, Kerala** **Sep 2020 – Mar 2021**

- Investigated the use of coir fibres as sustainable reinforcement to enhance concrete performance.
- Designed, cast, and tested CRC cube specimens, comparing results against conventional mixes.
- Analyzed mechanical behaviour and compiled findings highlighting coir's potential in low-cost eco-friendly construction.

INTERNSHIP

Civil Engineering Intern | **Kerala Public Works Department** | **Kasaragod, Kerala** **Jul 2018 – Jul 2018**

- Oversaw and supported RCC roof-slab construction, strengthening understanding of concrete behaviour, field practices, and quality control fundamentals.
- Verified shuttering, bar bending, and mix proportions during site activities, gaining hands-on exposure to material checks, measurement interpretation, and IS-code–based standards.
- Collaborated with site engineers and contractors to understand workflow sequencing, safety considerations, and decision-making in a live project environment.

SKILLS AND EXPERTISE

Modeling & Analysis | HEC-RAS (1D/2D), HEC-HMS, Visual MODFLOW Flex
GIS & Remote Sensing | QGIS, ArcGIS (basic), DEM Processing
Engineering Tools | AutoCAD, Excel
Domain Knowledge | Water Quality, Hydrology, Groundwater Flow, Environmental Monitoring

COURSEWORK

Semester 1 | Surface Water Hydrology & Hydrologic Systems, Advanced Fluid Mechanics, Remote Sensing & GIS Applications in Water Resources Engineering, Statistical Methods in Hydrology, Water Quality Modeling, Hydropower
Semester 2 | Soft Computing in Water Resources Engineering, Water Resources Systems Analysis and Design, Open-Source Scientific Programming for Water Resources Engineering, Urban Hydrology and Drainage, Computational Hydraulics and Hydrology, Flow and Transport in Porous Media

PROFESSIONAL MEMBERSHIPS

Student Member | **American Society of Civil Engineers (ASCE)** | **NIT Calicut** **Sep 2025 – Jul 2027**
Student Member | **Bureau of Indian Standards (BIS)** | **NIT Calicut** **Sep 2025 – Jul 2027**

CERTIFICATIONS

- **Open-Source Flood Modelling & Risk Assessment** | National Institute of Hydrology, Ministry of Jal Shakti **Sep 2025**
- **SAP MM (Materials Management)** | LinkedIn Learning **Jan 2025**
- **Quantity Surveying** | Bricklane India **Jan 2022**
- **Concrete Technology** | NPTEL **Oct 2019**